Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI & LLMs, Agentic AI, RAG, MLOps, Autonomous Anomaly Detection, Vector Embeddings, Predictive Modeling & Regression, Time-Series Forecasting (SARIMA), Dimensional Modeling (Star/Snowflake Schemas, SCD Type 1/2), Data Warehousing, ETL/ELT Pipeline Architecture, Regulatory Compliance Intelligence, Cost Optimization Analytics.

Programming Languages & Tools: AWS (S3, Redshift), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Apache Spark / PySpark, Python, Power BI (DAX), Tableau, Collibra, KIRO IDE, SQL, Git.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Business Intelligence & AI Analytics Engineer — **May 2022 – Present**

- **Architected end-to-end data pipelines and dimensional models** utilizing Azure, AWS (S3, Redshift), Snowflake, and Apache PySpark to crunch multi-terabyte regulatory and compliance datasets with 99.9% data availability SLAs.
- **Deployed autonomous agentic workflows and Generative AI** using KIRO IDE and prompt orchestration to automate complex data profiling, schema discovery, and regulatory classification, reducing manual delivery turnaround by 40-50%.
- **Engineered autonomous machine learning anomaly detection frameworks** achieving 95% precision in identifying schema variations, distribution drift, and ETL compliance discrepancies to safeguard downstream analytical integrity.
- **Developed executive BI dashboards and semantic models** in Power BI (DAX) and Tableau, translating complex transactional telemetry and Key Operating Metrics into strategic insights for senior leadership.
- **Automated data quality reconciliation firewalls and governance lineage** using Collibra and programmatic checks, reducing operational turnaround by over 80% and ensuring auditable data reliability for compliance programs.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Built and deployed predictive machine learning models** (XGBoost, Random Forests, Scikit-Learn) to analyze customer behavior patterns, delivering actionable decision signals that improved service reliability by 20%.
- **Architected advanced time-series forecasting models** (SARIMA) to predict operational demand, optimizing workforce capacity allocation and reducing resource response times by 20%.
- **Engineered automated testing and evaluation frameworks** for enterprise conversational AI platforms (SpectrumIQ), cutting manual QA turnaround by 50% and accelerating production release cycles.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Translated complex data challenges into scalable analytics** solutions by collaborating directly with cross-functional stakeholders to align reporting infrastructures with corporate strategy.
- **Designed scalable data warehouse architectures** incorporating Star/Snowflake schemas and SCD Type 1/2 methodologies to power self-service analytics and real-time operational decision support.
- **Established engineering standards and code quality best practices** for BI and analytics functions, conducting design reviews and mentoring junior team members on technical excellence.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.